For example, if we have a *three-shards-and-15-replicas-per-shard* strategy, with each shard sized at 50 GB, we must ensure that all 15 replicas have enough capacity not only to store documents on disk but also for heap memory:

- Shard memory: 3 × 50 GB/shard = 150 GB
- Replica memory per shard: 15 × 50 GB/replica = 750 GB/per shard
- (Replica memory for 3 shards = 3 × 750 GB = 2250 GB)
- Total memory for both shards and replicas on a given node = 150 GB + 750 GB = 900 GB
- (Grand total for 20 nodes = 18 TB)

A whopping 18 TB is required for one index with a *three-shards-and-15-replicas-per-shard* strategy. In addition to this initial disk space, we need further disk space to run the server smoothly. So, we must work through the capacity requirements judiciously. We discuss how to set up clusters with memory and disk space in chapters 14 and 15.

MULTI-NODE MULTICLUSTERS

We briefly looked at a node as an instance of an Elasticsearch server. As stated earlier, when we start the Elasticsearch application, we are essentially initializing a node. By default, this node joins a one-node cluster. We can create a cluster of any number of nodes based on our data requirements. We can also create multiclusters, as shown in figure 3.16, but doing so depends on an organization's use cases.

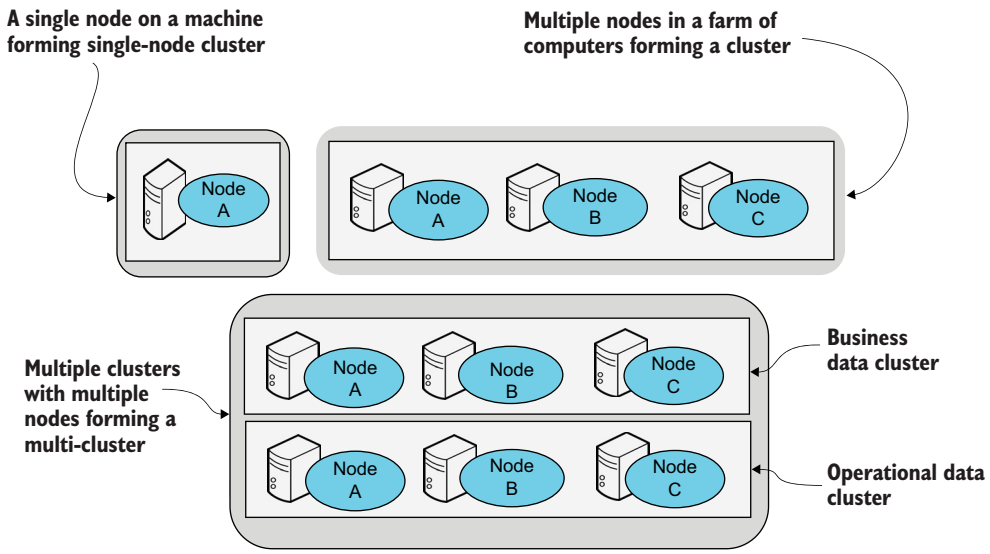


Figure 3.16 Varied cluster configurations